

Name: _____

Period: _____

Date: _____

Topic 1.4 Guided Notes: The Brain

Unit 1: Biological Bases of Behavior | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic1.4

Part 1: The Old Brain (Brainstem and Cerebellum)

Word Bank

<i>Cerebellum</i>	<i>Reticular Activating System</i>	<i>Pons</i>
<i>Medulla</i>	<i>Reward Center</i>	

Located at the very base of the skull where the spinal cord swells as it enters the brain, the brainstem is the oldest and most fundamental part of the central nervous system. It is responsible for automatic survival functions.

- _____ : located at the absolute base of the brainstem, this structure controls life-sustaining autonomic functions such as heartbeat, breathing, and blood pressure.
- _____ : sitting just above the medulla, this structure helps coordinate automatic movements and plays a critical role in regulating sleep and arousal by connecting the brainstem to the cerebellum.
- _____ : the "little brain" attached to the rear of the brainstem. It is essential for coordinating voluntary movement, maintaining balance and posture, and processing implicit (muscle) memories.
- _____ : a network of neurons that travels through the brainstem, playing an essential role in regulating alertness, attention, and sleep-wake cycles by filtering incoming sensory information.
- _____ : a group of brain structures, including areas of the limbic system, that are activated by pleasurable experiences and reinforce behaviors through dopamine release.

Apply It: Which Old Brain Structure?

Using the word bank above, identify which structure best matches each scenario.

#	Scenario	Structure
1	A patient can no longer maintain her balance or coordinate smooth, voluntary movements after an injury.	
2	A person in a coma shows no signs of arousal or wakefulness after this structure was severed.	
3	A rat with an electrode implanted in this structure will repeatedly press a lever to stimulate it, ignoring food and water.	

Part 2: The Emotional Center (The Limbic System)

Word Bank

<i>Thalamus</i>	<i>Pituitary Gland</i>	<i>Hypothalamus</i>
<i>Hippocampus</i>	<i>Amygdala</i>	

Sitting like a donut between the older brainstem and the newer cerebral hemispheres is the Limbic System. This neural system is heavily associated with intense emotions (like fear and aggression) and foundational drives (like hunger and sex).

- _____: two lima-bean-sized neural clusters that are heavily linked to raw emotions, particularly fear and aggression.
- _____: a seahorse-shaped structure critical for the processing and storage of explicit, conscious memories. If damaged, a person cannot form new factual memories (anterograde amnesia).
- _____: situated just below the thalamus, it directs maintenance activities (eating, drinking, body temperature), helps govern the endocrine system via the pituitary gland, and is linked to emotion and reward.
- _____: often called the body's "master gland," this small structure hangs just below the hypothalamus and takes its orders directly from it, releasing hormones that control other endocrine glands.
- _____: the brain's central sensory control center, located on top of the brainstem. It receives information from all the senses (except smell) and routes that data to the higher brain regions.

Apply It: Which Limbic Structure?

#	Scenario	Structure
1	After a head injury, a patient can no longer form any new long-term factual memories, though older memories remain intact.	
2	A patient's blood sugar, body temperature, and thirst regulation are all disrupted after a small structure below the thalamus is damaged.	
3	Stimulating a specific cluster of neurons in a lab animal's brain triggers sudden, intense aggression.	

Part 3: The New Brain (The Cerebral Cortex)

Word Bank

<i>Association Areas</i>	<i>Prefrontal Cortex</i>	<i>Higher-Order Thinking</i>
<i>Somatosensory Cortex</i>	<i>Corpus Callosum</i>	<i>Occipital Lobes</i>
<i>Executive Functioning</i>	<i>Temporal Lobes</i>	<i>Parietal Lobes</i>
<i>Motor Cortex</i>	<i>Frontal Lobes</i>	

The Cerebral Cortex is the intricate fabric of interconnected neural cells covering the cerebral hemispheres. It is divided into two hemispheres, connected by a thick band of neural fibers called the _____.

The Four Lobes

- _____: located directly behind your forehead. These lobes handle linguistic processing, _____ (complex reasoning like analysis, evaluation, and creative problem-solving), and _____ — the cognitive processes, including working memory, flexible thinking, and self-control, that let you plan ahead and resist impulses.

Most of this heavy lifting happens in the _____, the forward-most part of the frontal lobes responsible for judgment, planning, and processing new memories. The frontal lobes also contain the _____ at their rear, which controls most voluntary skeletal movement.

- _____: located at the top of the head toward the rear. They receive sensory input for touch, pressure, temperature, and body position through the _____, which registers and processes body touch and movement sensations. The rest is made up of _____ — regions responsible for higher mental functions like integrating sensory information into a coherent picture.
- _____: located at the back of the head. These lobes include areas that receive and process visual information from your eyes.
- _____: located roughly above the ears. They include the auditory areas, each receiving information primarily from the opposite ear. They also play a role in recognizing faces.

Apply It: Which Lobe?

#	Scenario	Lobe
1	A patient loses the ability to feel pressure and temperature on the left side of her body after a head injury.	
2	A patient can no longer plan ahead or control her impulses after a severe frontal injury.	
3	A patient can hear sounds perfectly but cannot recognize a familiar face by sight after a head injury.	
4	A patient loses part of his field of vision after damage to the back of his head.	

Part 4: Language Processing and Split-Brain Research

Word Bank

<i>Contralateral Organization</i>	<i>Split Brain Research</i>	<i>Left Hemisphere</i>
<i>Broca's Area</i>	<i>Wernicke's Area</i>	

While most functions are distributed across both hemispheres, language is typically lateralized to the _____. Two specific regions are heavily tested:

- _____: located in the left frontal lobe, it directs the muscle movements involved in speech production. (Damage causes expressive aphasia: you know what you want to say, but physically cannot produce the words.)
- _____: located in the left temporal lobe, it is involved in language comprehension. (Damage causes receptive aphasia: you can speak fluently, but the words make no logical sense — a "word salad.")

Apply It: Broca's or Wernicke's Aphasia?

#	Scenario	Area
1	A patient speaks in long, fluent sentences, but the words are strung together nonsensically and make no sense.	
2	A patient understands everything said to her perfectly but can only get out slow, halting, effortful single words.	

Evidence for this lateralization comes from _____: studies of individuals whose corpus callosum has been surgically severed, often to stop severe epilepsy. Researchers test for hemispheric specialization using the brain's _____ — the wiring quirk where the left hemisphere controls (and receives sensory information from) the right side of the body, and the right hemisphere handles the left.



Video Notes: Meet Your Master (Crash Course Psychology #4)

Answer the questions below as you watch — some ideas will already be familiar from Parts 1-3, and that's fine; reinforcing them is the point.

- According to the video, what happened to Phineas Gage, and what did his case reveal about the relationship between the brain and personality?
- What is the central nervous system, according to the video, and how does it relate to the peripheral nervous system?
- The video compares the brain to a set of Russian nesting dolls. What does this comparison mean about how the old brain and new brain developed?

- According to the video, what does the hippocampus do, and what happens if it's damaged?
- What is the difference between the motor cortex and the somatosensory cortex, according to the video?
- What are association areas, and how are they different from the motor or sensory cortex, according to the video?
- The video describes a "pop psychology" claim that creative people are "right-brained" and analytical people are "left-brained." According to the video, why is this claim misleading?

Part 5: The Adaptable Brain, and How We Study It

Word Bank

<i>Lesioning</i>	<i>Case Studies</i>	<i>fMRI</i>
<i>EEG</i>	<i>Neuroplasticity</i>	

_____ refers to the brain's ability to change, reorganize, and build new neural pathways throughout life, especially in response to learning, experience, or following injury. This is why children recover from severe brain injuries far better than adults.

How Do We Know? (Studying the Brain)

- _____: short for electroencephalogram, an amplified recording of the waves of electrical activity sweeping across the brain's surface, useful for studying sleep stages or catching a seizure as it happens.
- _____: a brain imaging technique that measures changes in blood flow to show which areas of the brain are active during a specific mental task.
- _____: in-depth investigations of a single individual, often someone with a rare or unusual brain injury.
- _____: studying the effects of damage to a specific brain structure, whether from natural injury or deliberately induced Lesions, to work backward and figure out what that region was responsible for.

6. Don't Trip Up! (Common Misconceptions)

The _____ Cortex is in the Frontal lobe (it sends signals out to move), while the _____ Cortex is right behind it in the Parietal lobe (it receives sensory signals in).

_____ 's area is for speaking (production). _____ 's area is for comprehending (understanding).

You do NOT only use 10% of your brain — brain scans show activity cascading across the entire brain even during simple tasks or sleep. You use 100% of your brain.

7. Quick Check: Review Questions

Circle the best answer for each question below.

1. A patient suffers severe damage to the very base of his brainstem, resulting in an inability to regulate his own heartbeat, breathing, and blood pressure without life support. Which structure was most likely damaged?

- A. Thalamus
- B. Amygdala
- C. Cerebellum
- D. Medulla

2. A man is bitten by a dog as a child and, from then on, experiences intense, automatic fear whenever he sees any dog, even friendly ones. Which limbic structure is most heavily linked to this raw fear response?
- A. Thalamus
 - B. Pons
 - C. Amygdala
 - D. Hippocampus
3. After a severe brain injury, a patient can still recall events and skills from before the accident but is completely unable to form any new factual memories going forward. Which structure was most likely damaged?
- A. Hippocampus
 - B. Cerebellum
 - C. Occipital Lobe
 - D. Amygdala
4. A structure sitting on top of the brainstem receives incoming information from all the senses except smell and routes that data to the appropriate higher brain regions for further processing. This structure is the:
- A. Amygdala
 - B. Pons
 - C. Thalamus
 - D. Hypothalamus
5. A patient can understand language perfectly and knows exactly what she wants to say, but after a stroke, she physically cannot produce coherent speech. Damage to which brain area, located in the left frontal lobe, best explains this deficit?
- A. The Somatosensory Cortex
 - B. The Occipital Lobe
 - C. Broca's Area
 - D. Wernicke's Area
6. A person who has been blind since birth shows unusually strong touch and hearing abilities. Brain scans reveal that her occipital lobe, normally dedicated to vision, has been recruited to help process this touch and sound information instead. This reorganization is best explained by:
- A. The 10% Myth
 - B. Neuroplasticity
 - C. The Reticular Activating System
 - D. Contralateral Organization



Check Your Progress

Done with your notes? Head over to the [Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>